

Clean Immediate-Opt-Out Baseline

A Formal Laboratory for Informational Power Through Pivotality

Working formal appendix

2026-08-04

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1 Status and scope

This document is the autonomous formal laboratory for Goal 1. It replaces the superseded delayed-continuation derivation with a clean benchmark in which

$$\pi_H = 0, \quad b_0 = b_1 = 0,$$

and a no vote by type θ of the hegemon is an immediate, irreversible opt-out. In Round 1 that action pays o_θ immediately and without discounting. It does not also give the hegemon a discounted continuation option.

The current version contains the complete Gate 0 contract. The independent read-only audit returned PASS without substantive reservation on candidate commit `fff4a35d1572c08bb2098cae9ad264a2eba80b41`. Analytical equilibrium results are not presumed in the Gate 0 specification; they must be derived separately by backward induction.

Current formal status: the extensive-form Gate 0 audit passed on the cited commit. Phase 1 has rederived the clean game under weak PBE and found endogenous existence restrictions, boundary multiplicity, and a conditional comparison domain. The analytical candidate awaits reproducible checks and final independent review; no protocol change has been introduced.

Manuscript boundary: `formal_model_v6.Rmd` and `formal_model_v5.Rmd` are reference objects only. This Goal does not edit either manuscript.

Excluded from the baseline: $y + b_\theta$, a rejection payoff βC_θ chosen by a no vote in Round 1, the hybrid $\max\{o_\theta, \beta C_\theta\}$, the decomposition $t_\theta = d_\theta - b_\theta$, positive recognition probability for H , endogenous rule choice, sequential roll-call voting, and the archived feasibility/C-B-R architecture. A discounted Round-2 payoff after H votes yes and weak votes make the Round-1 ballot fail is part of the baseline; it is not the payoff from opting out.

1.1 Provenance

The pre-reset state is commit `84a644128586d4f4d81c022cd7d3e09454ee8004`, recorded in the user-approved annotated tag `pre-clean-optout-goal1-2026-08-03`. The initial SHA-256 of `formal_model_v6.Rmd` is `f18a999300c88e32bdc7542f3249cd258afd19a0c251c0b29c2c387a7131dbc1`.

2 Gate 0: complete game contract

2.1 Players, type, prior, and parameters

There are $N \geq 3$ states: one hegemon H and $m = N - 1$ weak states. Nature draws

$$\theta \in \{0, 1\}$$

before Round 1. The type persists. Only H observes θ . Weak states share the prior

$$\mu = \Pr(\theta = 1) \in [0, 1].$$

The Bayesian derivations use $\mu \in (0, 1)$. The endpoints are treated as degenerate-belief cases or limits, and not as positive-probability separating histories containing both types.

The weak coalition has a fixed, known institutional surplus normalized to one. The private type does not change this pie. Weak outside payoffs are zero. The clean primitive domain is

$$0 \leq o_0 < o_1 \leq \bar{y} \leq 1, \quad \beta \in (0, 1]. \quad (\text{G0-domain})$$

If a current agreement includes H , its payoff is y . Thus $b_0 = b_1 = 0$. The outside payoff o_θ is external to the institutional pie. It is never subtracted from weak surplus and is never paid together with y in the same history.

The institutional rule $R \in \{U, M\}$ is fixed exogenously for each comparison. There is no endogenous rule-choice stage.

2.2 Common knowledge and solution concept

The primitives, parameters, type distribution, recognition rule, contract space, ballot protocol, quotas, implementation rule, payoff consequences, and tie-breaking rules stated here are common knowledge.

The solution concept is perfect Bayesian equilibrium (PBE). An assessment contains the weak states' collective formation strategy; every weak proposer's proposal strategy at each public history where it is recognized; every active non-proposer's behavioral voting strategy at each ballot information set; H 's type-contingent behavioral voting strategy; and a belief system at every information set where θ remains payoff-relevant. Strategies are sequentially rational given the belief system, and beliefs follow Bayes' rule wherever possible. Every result below must state whether it concerns all PBEs, pure-strategy PBEs, a constructed or selected PBE outcome, or a selection-free bound. The solution concept does not select among multiple PBEs unless one of the two stated tie rules applies.

2.3 Horizon, recognition, and time units

The game has exactly two bargaining rounds, R1 and R2. There is no R3. In each round that is reached, a new proposer is drawn publicly and independently of the previous recognition, the type, and the public history. The baseline fixes

$$\pi_H = 0$$

in every round. Each weak state is recognized with probability $1/m$, conditional on that round being reached.

Round 1 defines the payoff unit. A payoff realized in R2 is multiplied by β in Round-1 units. A no vote by H in R1 pays o_θ at once, without β . A no vote by H in R2 pays o_θ at the R2 date, or βo_θ in Round-1 units.

R2 is terminal. If H is still active and its R2 ballot fails, then H receives o_θ at the R2 date, regardless of whether it voted yes or no; weak states receive zero. This is the terminal outside payoff, not a second payment. If H opted out in R1, it receives o_θ exactly once and receives no R2 flow.

2.4 Formation

After Nature draws θ , but before the R1 recognition draw, the m weak states make one collective, all-or-nothing formation decision. They do not observe θ . The hegemon observes its type but takes no formation action.

If the institution forms, every weak state pays an external sunk cost $\chi \geq 0$. The cost does not reduce the unit pie. If the institution does not form, every weak state receives zero and H receives o_θ .

For rule R , the gross payoff relevant to collective formation is

$$V_W^R(\mu) = \frac{1}{m} \mathbb{E} \left[\sum_{j=1}^m u_j^R \mid \text{formation} \right]. \quad (\text{G0-entry-value})$$

The expectation integrates over the type, recognition, public histories, and any symmetric selection among weak coalition partners. Formation occurs iff $V_W^R(\mu) - \chi \geq 0$; equality leads to formation. This collective primitive is not an assertion that every realized weak state individually prefers formation under an asymmetric coalition selection.

2.5 Contract space and budget identities

Suppose weak state i is proposer. While H is active, a proposal is

$$s_i = (y, (x_j)_{j \neq i}),$$

where

$$y \in [0, \bar{y}], \quad x_j \geq 0,$$

and the proposer's residual if H participates is

$$x_i^H = 1 - y - \sum_{j \neq i} x_j \geq 0. \quad (\text{G0-H-budget})$$

The same space applies in R1 and R2 and under both rules. The component y is a public offer conditional on the observed participation of H ; there is no hidden inclusion indicator. If H votes yes and the proposal passes, it is included and receives y . Every named weak-state offer x_j is paid when the proposal passes, independently of that recipient's individual vote.

If H votes no under majority and the weak votes nevertheless reach the quota, the agreement passes without H . The conditional component y is not burned and is not paid to H . It is reabsorbed by the weak proposer, so

$$x_i^{WO} = 1 - \sum_{j \neq i} x_j = x_i^H + y. \quad (\text{G0-WO-budget})$$

The hegemon receives o_θ externally. Hence both implemented budget identities close at one:

$$y + \sum_{j \neq i} x_j + x_i^H = 1, \quad \sum_{j \neq i} x_j + x_i^{WO} = 1. \quad (\text{G0-budget-closure})$$

After an earlier opt-out, only the m weak states remain active, every proposal fixes $y = 0$, and the weak-only identity in (G0-WO-budget) applies. If a ballot fails, no current x_j and no current y is implemented.

These rules make y contingent only on publicly observed participation, not on the private type.

2.6 Common simultaneous ballot

The game is sequential and public across rounds: proposal, ballot, publication of the complete vote vector and outcome, followed by any continuation. It is not sequential within a ballot.

The proposer is counted as voting yes. All other active players observe the proposal and public history and vote simultaneously. The individual votes are revealed only after every ballot has been cast. The same ballot protocol is used under both rules. Only the quota changes:

$$Q_U = N, \quad Q_M = q = \left\lfloor \frac{N}{2} \right\rfloor + 1. \quad (\text{G0-quotas})$$

Because $N \geq 3$, $q \leq m$. While H remains active, the electorate contains all N states. After opt-out, the active electorate contains only the m weak states, but the majority quota remains the original q ; it is not recalculated. The unanimity quota remains N . Therefore an opt-out makes future unanimity impossible and that branch ends immediately.

Players vote yes when indifferent. Strategies are functions of information available before the ballot closes. No strategy may condition on a simultaneous vote that has not yet been observed.

2.7 Immediate opt-out

A no vote by type θ of H is an immediate and irreversible opt-out. No current or future agreement can then include H . In R1 the action pays o_θ immediately. In R2 it pays o_θ at the R2 date. There is no additional choice between opt-out and a discounted continuation.

Under unanimity, an R1 no vote by H makes the quota impossible and ends the branch. Under majority, if the weak yes votes already reach q , their current weak-only agreement passes. Otherwise the game proceeds to an R2 weak-only subgame with the original quota q . A joint R1 rejection by H and weak voters follows the same rules.

A yes vote by H is not an opt-out. If the R1 ballot then fails only because weak votes do not reach the quota, H remains active and the game proceeds to R2 under the same institutional rule.

2.8 Exhaustive voting-history table

Let $z \in \{1, \dots, m\}$ denote the number of weak yes votes, including the weak proposer's automatic yes vote. The tables in this section are generated from the single versioned tab-separated source `tables/clean_optout_gate0_histories_piH0.` HTML displays the complete 19-column table in a scroll box. To preserve legibility, PDF displays the same rows and columns as four keyed panels. The concise entries use `C_H2^R(theta,h2)` only for the legitimate continuation after an R1 yes vote by H followed by a weak-caused failure. They never use that object as the payoff from a no vote.

Table 1: Gate 0 histories, panel A: timing, electorate, quota, and votes.

history_id	round	rule	active players	quota	quota reached?	H vote
G01	R1	U	H plus m weak	N	yes: $z=m$	yes
G02	R1	U	H plus m weak	N	no: $z<m$	yes
G03	R1	U	H plus m weak	N	no: $z=m$	no
G04	R1	U	H plus m weak	N	no: $z<m$	no
G05	R1	M	H plus m weak	q	yes: $z \geq q-1$	yes
G06	R1	M	H plus m weak	q	no: $z \leq q-2$	yes
G07	R1	M	H plus m weak	q	yes: $z \geq q$	no
G08	R1	M	H plus m weak	q	no: $z=q-1$	no
G09	R1	M	H plus m weak	q	no: $z \leq q-2$	no
G10	R2	U	H plus m weak	N	yes: $z=m$	yes
G11	R2	U	H plus m weak	N	no: $z<m$	yes
G12	R2	U	H plus m weak	N	no: $z=m$	no
G13	R2	U	H plus m weak	N	no: $z<m$	no
G14	R2	M	H plus m weak	q	yes: $z \geq q-1$	yes
G15	R2	M	H plus m weak	q	no: $z \leq q-2$	yes
G16	R2	M	H plus m weak	q	yes: $z \geq q$	no
G17	R2	M	H plus m weak	q	no: $z=q-1$	no
G18	R2	M	H plus m weak	q	no: $z \leq q-2$	no
G19	R2	M	m weak; H already opted out	original q	yes: $z \geq q$	not applicable
G20	R2	M	m weak; H already opted out	original q	no: $z < q$	not applicable
G21	R2	U	m weak; H already opted out	N	structurally impossible	not applicable

Table 2: Gate 0 histories, panel B: implementation and the disposition of the participation package.

history_id	participation offer to H?	failure cause	H included?	H remains?	y implemented?	destination of y
G01	yes	none	yes	no continuation	yes	paid to H
G02	yes	weak-vote shortfall	no	yes	no	not implemented
G03	yes	H-no is decisive	no	no	no	not implemented
G04	yes	joint H-no and weak rejection	no	no	no	not implemented
G05	yes	none	yes	no continuation	yes	paid to H
G06	yes	weak-vote shortfall	no	yes	no	not implemented
G07	yes	none; weak votes reach q	no	no	no	reabsorbed by proposer
G08	yes	H-no is decisive for failure	no	no	no	not implemented
G09	yes	joint H-no and weak shortfall	no	no	no	not implemented
G10	yes	none	yes	no continuation	yes	paid to H
G11	yes	weak-vote shortfall	no	no	no	not implemented
G12	yes	H-no is decisive	no	no	no	not implemented
G13	yes	joint H-no and weak rejection	no	no	no	not implemented
G14	yes	none	yes	no continuation	yes	paid to H
G15	yes	weak-vote shortfall	no	no	no	not implemented
G16	yes	none; weak votes reach q	no	no	no	reabsorbed by proposer
G17	yes	H-no is decisive for failure	no	no	no	not implemented
G18	yes	joint H-no and weak shortfall	no	no	no	not implemented
G19	no; y fixed at 0	none	no	no	no; fixed at 0	not available
G20	no; y fixed at 0	weak-vote shortfall	no	no	no; fixed at 0	not available
G21	no; y fixed at 0	H absent, so unanimity quota cannot be met	no	no	no; fixed at 0	not available

Table 3: Gate 0 histories, panel C: payoffs and time units.

history_id	H payoff	weak payoffs	discount factor
G01	y	x_j ; proposer gets $1-y-\text{sum}(x_j)$	1
G02	$\beta * C_{H2}^U(\theta, h_2)$	β times R2-U continuation payoffs	β
G03	o_θ	0	1
G04	o_θ	0	1
G05	y	x_j ; proposer gets $1-y-\text{sum}(x_j)$	1
G06	$\beta * C_{H2}^M(\theta, h_2)$	β times R2-M continuation payoffs with H active	β
G07	o_θ	x_j ; proposer gets $1-\text{sum}(x_j)$	1
G08	o_θ	β times R2 weak-only continuation payoffs	β for weak continuation; H paid at R1
G09	o_θ	β times R2 weak-only continuation payoffs	β for weak continuation; H paid at R1
G10	$\beta * y$	β times x_j ; proposer gets $\beta * [1-y-\text{sum}(x_j)]$	β
G11	$\beta * o_\theta$	0	β
G12	$\beta * o_\theta$	0	β
G13	$\beta * o_\theta$	0	β
G14	$\beta * y$	β times x_j ; proposer gets $\beta * [1-y-\text{sum}(x_j)]$	β
G15	$\beta * o_\theta$	0	β
G16	$\beta * o_\theta$	β times x_j ; proposer gets $\beta * [1-\text{sum}(x_j)]$	β
G17	$\beta * o_\theta$	0	β
G18	$\beta * o_\theta$	0	β
G19	0 additional	β times x_j ; proposer gets $\beta * [1-\text{sum}(x_j)]$	β
G20	0 additional	0	β
G21	0 additional	0	not applicable

Table 4: Gate 0 histories, panel D: beliefs and continuation subgames.

history_id	relevant posterior	terminal or continuation	next subgame
G01	irrelevant after agreement	terminal	none
G02	nu(h2) from proposal and full vote vector; continuation may depend on full h2	continuation	R2 U with H active
G03	irrelevant after opt-out	terminal	none
G04	irrelevant after opt-out	terminal	none
G05	irrelevant after agreement	terminal	none
G06	nu(h2) from proposal and full vote vector; continuation may depend on full h2	continuation	R2 M with H active
G07	irrelevant after opt-out	terminal	none
G08	irrelevant after opt-out	continuation	R2 M weak-only
G09	irrelevant after opt-out	continuation	R2 M weak-only
G10	irrelevant after terminal agreement	terminal	none
G11	irrelevant at terminal date	terminal	none
G12	irrelevant at terminal date	terminal	none
G13	irrelevant at terminal date	terminal	none
G14	irrelevant after terminal agreement	terminal	none
G15	irrelevant at terminal date	terminal	none
G16	irrelevant at terminal date	terminal	none
G17	irrelevant at terminal date	terminal	none
G18	irrelevant at terminal date	terminal	none
G19	theta is payoff-irrelevant	terminal	none
G20	theta is payoff-irrelevant	terminal	none
G21	theta is payoff-irrelevant	unreachable; branch ended at opt-out	none

The R2-unanimity row with H already absent is included to document a structurally unreachable class: the branch ends when the opt-out occurs because the original quota N cannot be met by m active weak states. It is not a latent R2 action.

3 Information, beliefs, and incentive constraints

3.1 Public histories and beliefs

Let h denote the public history before a proposal, let s_i be the public proposal, and let

$$I = (h, s_i)$$

denote a ballot information set before simultaneous votes are cast. After the ballot, the public history is

$$h_2 = (h, s_i, a_H, v_W, \text{quota outcome, opt-out status}).$$

Thus continuation strategies may condition on the full proposal and revealed vote vector, not only on the posterior. Let $\mu(h)$ be the belief before the ballot. Whenever the denominator is positive, Bayes' rule requires

$$\nu(h_2) = \frac{\mu(h) \Pr(h_2 \mid \theta = 1, h)}{\mu(h) \Pr(h_2 \mid \theta = 1, h) + \{1 - \mu(h)\} \Pr(h_2 \mid \theta = 0, h)}. \quad (\text{G0-Bayes})$$

The posterior $\nu(h_2)$ and a continuation payoff $C_{H,2}^R(\theta, h_2)$ are distinct objects. Histories with identical posteriors can induce different continuation strategies and payoffs.

The baseline uses the expression **weak-vote-passive assessment**. It is a maintained assessment of the simultaneous public ballot, not a refinement, D1, the Intuitive Criterion, a sequential-equilibrium restriction, or a characterization of all PBEs.

A unilateral vote deviation by a weak state is not treated as direct evidence about θ , because weak states do not observe θ . Following a weak-voter deviation, the posterior is the posterior implied by the public proposal and H 's simultaneous type-contingent strategy and vote, not by the action of an uninformed deviator. A separating vote by H can update beliefs when the continuation keeps H active and the type remains payoff-relevant.

Every zero-probability continuation in which H remains active must state the required off-path posterior and identify which result depends on it. Off-path posteriors are components of an assessment, not primitives of the game. They cannot be selected to delete an inconvenient branch. Once H opts out, θ is payoff-irrelevant for the weak-only continuation and no posterior is needed to solve it.

3.2 Hegemon's expected ballot IC

Fix an R1 ballot information set $I = (h, s_i)$ and type θ . Let v_W be the simultaneous vector of non-proposer weak votes, let $A_R^Y(v_W)$ indicate that the rule- R quota is reached when H votes yes, and let $\sigma_W(v_W \mid I)$ be the probability induced by the weak strategies. The correct R1 payoff from voting yes is

$$EU_H^R(Y \mid \theta, I) = \sum_{v_W} \sigma_W(v_W \mid I) \left[A_R^Y(v_W) y + \{1 - A_R^Y(v_W)\} \beta C_{H,2}^R(\theta, h_2(h, s_i, Y, v_W)) \right]. \quad (\text{G0-H-yes-R1})$$

The payoff from voting no is always

$$EU_H^R(N \mid \theta, I) = o_\theta. \quad (\text{G0-H-no-R1})$$

This equality holds whether a majority weak-only agreement passes immediately or the weak states continue alone: H has opted out and receives no further flow. Sequential rationality requires

$$EU_H^R(Y \mid \theta, I) \geq o_\theta \quad (\text{G0-H-IC-R1})$$

for a prescribed yes vote, with the reverse weak inequality for a prescribed no vote.

At an R2 ballot information set $I_2 = (h_2, s_i)$, a failed ballot with H still active pays the terminal outside payoff. Thus, in Round-1 units,

$$EU_H^R(Y \mid \theta, I_2) = \beta \sum_{v_W} \sigma_W(v_W \mid I_2) [A_R^Y(v_W)y + \{1 - A_R^Y(v_W)\}o_\theta], \quad EU_H^R(N \mid \theta, I_2) = \beta o_\theta. \quad (\text{G0-H-IC-R2})$$

At an R1 information set where weak strategies make $A_R^Y(v_W) = 1$ with probability one, the comparison directly reduces to

$$y \geq o_\theta. \quad (\text{G0-direct-cutoff})$$

Therefore the direct threshold $y_\theta^* = o_\theta$ is derived at a certain-implementation information set. It is not a global voting rule. If weak-caused failure has positive probability, equation (G0-H-IC-R1) must integrate current implementation and the legitimate R2 continuation after a yes vote.

R2 has a special terminal algebra that does not change this scope statement. If

$$p_R(I_2) = \Pr(A_R^Y(v_W) = 1 \mid I_2),$$

then equation (G0-H-IC-R2) implies

$$EU_H^R(Y \mid \theta, I_2) - EU_H^R(N \mid \theta, I_2) = \beta p_R(I_2)(y - o_\theta). \quad (\text{G0-H-R2-difference})$$

Consequently, every $p_R(I_2) > 0$ produces the same sign as $y - o_\theta$, while $p_R(I_2) = 0$ makes H indifferent for every y and the voting tie rule prescribes yes. This is a consequence of terminal failure payoffs, not a global primitive cutoff and not a license to replace the R1 expected IC when failure has positive probability.

3.3 Weak-voter approval IC

Fix a non-proposing weak voter j at ballot information set $I = (h, s_i)$. Its strategy is chosen before the simultaneous ballots are revealed. Let $u_j(a_j, v_{-j}; I)$ be the payoff specified by the history table: an implemented x_j if the current agreement passes, the appropriate discounted continuation if R1 fails, or zero after terminal failure. Its approval condition is

$$\sum_{v_{-j}} \Pr(v_{-j} \mid I) u_j(Y, v_{-j}; I) \geq \sum_{v_{-j}} \Pr(v_{-j} \mid I) u_j(N, v_{-j}; I). \quad (\text{G0-W-IC})$$

The expectation includes H 's type-contingent simultaneous vote. A voter cannot condition on that vote before the ballot closes. If voter j is pivotal with probability one, equation (G0-W-IC) reduces to the usual comparison between x_j and the continuation induced by a no vote. If its vote never changes implementation or continuation, it is indifferent and votes yes. Offers are implemented for named recipients whenever the agreement passes, even if the recipient voted no.

3.4 Tie-breaking

Every player votes yes when indifferent. Among proposals that maximize the weak proposer's payoff, the selected proposal minimizes H 's expected payoff. No third tie-break is available. If two proposals tie for both the weak proposer and H but imply different downstream objects, the result must remain set-valued or exclude that boundary by a stated strict theorem domain; the proof may not select one silently.

4 Gate 0 audit checklist

The implementer records the following mechanical assessment, which does not substitute for the independent verdict.

Requirement	Candidate evidence	Implementer status
Every required history has a payoff and continuation	Versioned history tables, rows G01–G21	checked
No double payment to H	Contract identities and all H-no rows	checked
No reinclusion after opt-out	Rows G03, G04, G07–G09, G12, G13, and G16–G21	checked
No discounted continuation as payoff from H-no in R1	Equation (G0-H-no-R1) and table H-no rows	checked
Same simultaneous ballot under U and M	Equation (G0-quotas)	checked
Post-opt-out electorate and quotas defined	Common-ballot section and rows G19–G21	checked
H-including and weak-only budgets close	Equations (G0-H-budget)–(G0-budget-closure)	checked
H IC integrates simultaneous weak votes	Equations (G0-H-yes-R1)–(G0-H-IC-R2)	checked
Beliefs stated whenever type remains relevant	Beliefs section and G02/G06	checked
No equilibrium result presumed as a primitive	Status, primitives, and proof ledger	checked

Independent Gate 0 verdict: PASS without substantive reservation on commit `fff4a35`. The read-only reviewer found no critical, major, or minor substantive issue after the repair round. Gate 0 fixes the game; it does not presume that a PBE exists for every parameter vector. The backward-induction results below therefore leave Gate 0 closed while reporting endogenous existence restrictions faithfully.

5 Analytical rederivation

5.1 Belief discipline and theorem domains

The distinction between an on-path and a globally off-path proposal is essential. A proposal made with positive probability by an uninformed weak proposer preserves the current belief and subsequent positive-probability histories obey Bayes’ rule. After a globally zero-probability proposal, the posterior is a component of the PBE assessment and must be declared. The weak-vote-passive assessment prevents an uninformed unilateral vote deviation from being treated as a direct signal, but it does not silently turn weak PBE into sequential equilibrium.

All payoffs from deviations are evaluated under the true prior μ , even when voters at an off-path ballot hold a declared posterior ν . This separation is what generates the security-value discontinuities below.

The main theorems use the **regular interior domain**

$$\mu \in (0, 1), \quad \beta \in (0, 1), \quad 0 < o_0 < o_1 < 1. \quad (\text{regular-domain})$$

The loci $o_0 = 0$, $o_1 = 1$, $\beta = 1$, and degenerate priors are stated separately. They are not imported into the regular theorem by continuity.

5.2 Round 2 under unanimity

In terminal R2 unanimity, every non-proposing weak voter approves. A unilateral no either leaves failure unchanged or turns failure into an agreement paying that voter a nonnegative offer; equality invokes the yes-vote tie rule. Hence

H is pivotal and the R2 IC in (G0-H-R2-difference) yields the direct cutoff $y \geq o_\theta$.

At posterior ν , define

$$P = 1 - o_1, \quad L(\nu) = (1 - \nu)(1 - o_0), \quad S(\nu) = \max\{P, L(\nu)\}. \quad (\text{R2-U-value})$$

A pooling proposal sets $y = o_1$ and leaves weak total P . A low-only proposal sets $y = o_0$ and leaves expected weak total $L(\nu)$. At equality, the proposal tie-break selects low-only because it gives H the lower expected payoff. Thus

$$\nu_2^* = \frac{o_1 - o_0}{1 - o_0} \quad (\text{R2-U-cutoff})$$

is the terminal belief cutoff, the total weak value is $S(\nu)$, and each weak state's ex ante R2 value is $S(\nu)/m$. This is an all-PBE payoff result for the terminal subgame; allocation among weak states can still depend on the recognition draw.

5.3 Round 2 under majority

In terminal R2 majority, weak approval is weakly dominant: failure pays zero, an implemented named offer is nonnegative, and ties go to yes. All weak states therefore approve. A weak proposer sets all named weak offers to zero, the proposal passes even after H -no, and the proposer retains the unit pie. The conclusion is the same in the active- H and post-opt-out weak-only subgames:

$$C_{W,2}^M = \frac{1}{m}, \quad c := \beta C_{W,2}^M = \frac{\beta}{m}, \quad C_{H,2}^M(\theta) = o_\theta. \quad (\text{R2-M-values})$$

The outside payoff remains external to the institutional pie. This proposition is selection-free.

6 Round 1 under unanimity

6.1 On-path candidates

For the regular domain, write

$$D_0 = 1 - o_0, \quad L = (1 - \mu)D_0, \quad \delta = \frac{\beta(m-1)}{m}, \quad a = 1 - \delta. \quad (\text{U-notation})$$

If pooling is on path, a weak deviation after H -yes preserves posterior μ . Each non-proposer must receive $\beta S(\mu)/m$, and the maximal proposer payoff is

$$\pi_P = P - \delta S(\mu). \quad (\text{U-onpath-P})$$

If low-only is on path, H -yes reveals the low type. Each non-proposer must receive $\beta D_0/m$, so

$$\pi_L = aL. \quad (\text{U-onpath-L})$$

There is no high-branch continuation term in π_L : high-type H -no is immediate opt-out and ends unanimity. For $o_0 > 0$ and $\beta < 1$, an on-path weak-caused failure cannot be sustained because both types prefer immediate opt-out once Bayes disciplines the continuation.

6.2 Off-path completion lemma

Assume first that $\beta o_1 \geq o_0$, and define

$$d = \frac{\beta P}{m}, \quad G_L = (1 - \mu)(D_0 - \delta P), \quad G_P = aP. \quad (\text{U-security})$$

The declared off-path continuation after low-type H -yes can put enough mass on the high type to select terminal pooling. The low type then obtains βo_1 after weak-caused failure and is willing to vote yes; high-type H strictly opts out. This gives the following exhaustive completion.

1. If $y < o_0$, prescribe all weak voters yes and both types of H no. The proposer receives zero.
2. If $o_0 \leq y < o_1$ and every $x_j \geq d$, prescribe all weak voters yes, low-type H yes, and high-type H no. A pooling posterior after a unilateral weak deviation makes d the exact approval price. The best such deviation gives the proposer G_L .
3. If $y \geq o_1$ and every $x_j \geq d$, prescribe all weak voters and both types of H yes. The best such deviation gives G_P .
4. If some $x_j < d$, designate one underpaid weak rejector, prescribe low-type H yes and high-type H no, and use the pooling continuation. The rejector strictly prefers continuation. The proposer's payoff is at most $(1 - \mu)\beta P/m < G_L$.

Two or more weak no votes cannot occur because each rejector would be nonpivotal and the tie rule would require yes. A nondegenerate ballot mixture cannot enlarge the set: mixing requires indifference, where the same rule assigns yes with probability one.

If $\beta o_1 < o_0$, step 4 is impossible. The feasible proposal $y = o_1, x_j = 0$ then has no sequentially rational ballot completion: all weak yes makes each weak voter prefer the positive R2 continuation, two weak no votes violate the tie rule, and with exactly one weak no both types of H opt out. Thus the full game has no PBE on that open subdomain.

6.3 Unanimity existence theorem

Suppose (regular-domain) holds. Define

$$D_U = D_0 - \delta P > 0, \quad \mu_E = 1 - \frac{aP}{D_U}. \quad (\text{U-existence-cutoff})$$

Then a global unanimity PBE exists if and only if

$$\boxed{\beta o_1 \geq o_0 \quad \text{and} \quad G_P > G_L} \quad (\text{U-existence})$$

or, equivalently, $\beta o_1 \geq o_0$ and $\mu > \mu_E$.

Proof. The completion lemma shows that the proposer can force both G_L and G_P , while every proposal can be completed at no more than their maximum. Yet the on-path low-only price is $\beta D_0/m > d$, so

$$G_L - \pi_L = (1 - \mu)\delta(D_0 - P) > 0. \quad (\text{U-shaving-gap})$$

Low-only can therefore never be a regular PBE: the proposer shaves every weak offer to d when the deviation is off path, but Bayes would restore the higher price if that proposal became on path. Pooling can attain its off-path security payoff only when $P > L$, in which case $\pi_P = aP = G_P$. Direct algebra gives

$$\mu_E - \nu_2^* = \frac{\delta P(D_0 - P)}{D_0 D_U} > 0, \quad (\text{U-cutoff-order})$$

so $G_P > G_L$ already implies $P > L$. If equality holds, the intermediate deviation ties the proposer payoff and pays less to H ; the proposal tie-break selects it, but its on-path approval price jumps back to $\beta D_0/m$. No fixed

point exists at equality. Under the strict inequality, pooling strictly dominates every completed deviation and the completion lemma constructs a global PBE. $\square$

Every regular unanimity PBE in the existence domain has the same on-path outcome:

$$y = o_1, \quad T_W^U = P, \quad V_W^U = \frac{P}{m}, \quad E[u_H^U] = o_1. \quad (\text{U-regular-payoffs})$$

This is a pooling result, not a resurrection of the historical P/L/R characterization and not a claim about parameter vectors outside (U-existence).

6.4 Unanimity boundary results

The boundary results are separate propositions obtained from the same ballot best-response enumeration.

6.4.1 Zero low outside option

Let $\beta < 1$, $o_0 = 0$, and $o_1 < 1$. A global PBE always exists. If $L \geq P$, low-only is the only minimal total-payoff class and $T_W^U = L$. If $P > L$, canonical pooling exists, while low-only also exists if and only if

$$P - \delta \leq aL. \quad (\text{U-zero-low-multiplicity})$$

At equality the proposal tie-break favors low-only. Pooling assessments with a larger concession $y \in [o_1, \bar{y}]$ can also be sustained. If $T = 1 - y$, their maximal proposer payoff is $T - \delta P$. The strict interior of the constructed set satisfies

$$T \in [1 - \bar{y}, P], \quad T - \delta P > \max\{aL, P - \delta\}. \quad (\text{U-zero-low-overpay})$$

Equality is selected against pooling when the binding deviation pays less to H ; if both proposer and H payoffs tie, the outcome remains set-valued. Thus $S(\mu)/m$ is an open endpoint, not an all-PBE R1 value.

6.4.2 No terminal weak surplus

If $\beta < 1$, $o_0 > 0$, $o_1 = 1$, and $\mu \in (0, 1)$, no PBE exists. An off-path posterior selecting zero-surplus pooling lets the proposer shave the low-only approval price below the Bayes-consistent on-path price.

6.4.3 No discounting

If $\beta = 1$, rejection/continuation is admissible and gives proposer payoff $S(\mu)/m$. When $L \geq P$, low-only and rejection survive. When $P > L$, pooling and rejection survive; low-only additionally survives iff

$$P - \frac{(m-1)(1-o_0)}{m} \leq \frac{L}{m}. \quad (\text{U-beta-one-low})$$

Overpaid pooling and proposal mixing can add payoff-equivalent or lower-total outcomes. No timing uniqueness is claimed. This is the only baseline locus on which the historical rejection/delay label survives on path.

7 Round 1 under majority

7.1 Voting reduction and security value

For the regular domain, define

$$q = \left\lfloor \frac{N}{2} \right\rfloor + 1, \quad k = q - 1, \quad c = \frac{\beta}{m}, \quad (\text{M-notation})$$

and

$$E = 1 - kc, \quad B_M = (1 - \mu)(1 - o_0) + \mu c, \quad P = 1 - o_1, \quad F_M = \max\{E, B_M, P\}. \quad (\text{M-security})$$

For any non-proposing weak voter,

$$EU_j(Y) - EU_j(N) = (x_j - c) \Pr(j \text{ is pivotal}). \quad (\text{M-weak-IC})$$

If $z \geq q - 1$, H -yes implements and type θ approves iff $y \geq o_\theta$. If $z \leq q - 2$, H -yes still fails and pays $\beta o_\theta < o_\theta$; both types opt out. Outside all-weak-yes, the only regular pure ballot size is therefore $z = q - 1$.

The security value F_M is exact. Let $X = \sum_{j \neq i} x_j$, and let r count non-proposers offered at least c .

1. If $y < o_0$ and $r \leq k - 1$, a $z = q - 1$ rejection completion pays the proposer c ; if $r \geq k$, weak-only passage gives at most $1 - X \leq E$.
2. If $o_0 \leq y < o_1$, declare the off-path belief $\nu = 1$. With $r \leq k - 1$, choose $z = q - 1$; the true low state passes, the true high state continues, and the proposer receives at most B_M . With $r \geq k$, weak-only passage is forced and the proposer receives less than E .
3. If $y \geq o_1$, all weak voters are nonpivotal when all vote yes, and the proposer receives at most P .

Conversely, paying c to k partners guarantees E ; the proposal $y = o_1, x = 0$ guarantees P ; and $y = o_0, x = 0$ guarantees at least B_M . Hence no proposal system can give the proposer less than F_M , and off-path completions can hold every deviation to at most F_M .

7.2 Majority characterization by group size

7.2.1 Three states

For $N = 3$, the exact proposer payoff is

$$V_{\text{prop}}^M = F_M. \quad (\text{M-N3-value})$$

Exclusion is selected when E maximizes; low-only is selected when B_M maximizes; pooling requires $P > \max\{E, B_M\}$. A tie $E = B_M$ remains set-valued because both classes give H exactly its outside payoff. Their weak totals are

$$T_W^E = 1, \quad T_W^B = (1 - \mu)(1 - o_0) + \mu\beta, \quad T_W^P = P. \quad (\text{M-N3-totals})$$

7.2.2 Four states

For $N = 4$, the deviation $y = o_0, x = 0$ guarantees B_M when off path but has no Bayes-consistent pure ballot on path. The only on-path classes that can attain the security value are exclusion and pooling. Therefore

$$\boxed{\text{a majority PBE exists iff } E \geq B_M \text{ or } P > B_M} \quad (\text{M-N4-existence})$$

or, equivalently, it fails to exist iff $B_M > E$ and $B_M \geq P$. Exclusion is selected iff $E \geq \max\{B_M, P\}$, and pooling iff $P > \max\{B_M, E\}$. The boundary $B_M = P > E$ also has no PBE: the unattainable deviation ties pooling in proposer payoff and the tie-break favors its lower payoff to H .

7.2.3 Five or more states

For $N \geq 5$, all-weak-yes is oversized. A majority PBE always exists, and the exact set of proposer payoffs is

$$V_{\text{prop}}^M \in [F_M, 1]. \quad (\text{M-large-value})$$

Every regular PBE belongs to one of three classes.

1. **Exclusion:** $y < o_0$, with weak total $T_W^M = 1$ and H receiving o_θ . Redundant weak transfers support proposer payoffs throughout $[F_M, 1]$.
2. **Separating current passage:** $o_0 \leq y < o_1$, low-type H votes yes and high-type H votes no, but the oversized weak coalition passes the agreement in both states. If the total named offers are X ,

$$v = 1 - X - (1 - \mu)y \geq F_M, \quad T_W^M = 1 - (1 - \mu)y, \quad E[u_H^M] = (1 - \mu)y + \mu o_1. \quad (\text{M-large-separating})$$

The canonical $y = o_0, X = 0$ outcome exists iff $1 - (1 - \mu)o_0 \geq E$. At a proposer-payoff tie, a positive rent $y - o_0$ survives only when it also satisfies the stated proposal tie-break.

3. **Pooling:** $y = o_1, X = 0$, which exists iff $P > \max\{E, B_M\}$. It gives $T_W^M = P$ and $E[u_H^M] = o_1$.

Low-only with delay is not a regular large- N PBE because its maximal proposer payoff is strictly below B_M . Across all three classes,

$$F_M \leq T_W^M \leq 1, \quad \frac{F_M}{m} \leq V_W^M \leq \frac{1}{m}. \quad (\text{M-weak-bounds})$$

The same weak-total bounds hold for $N = 3$ and for every existing $N = 4$ PBE, using the class-specific totals above.

7.3 No-Cheap-H changes

The historical condition $o_0 \geq c$ only compares exclusion with a minimal-coalition low-only proposal. Under weak PBE, the binding off-path comparison is B_M versus E :

$$B_M - E = kc - o_0 + \mu(o_0 + c - 1). \quad (\text{M-no-cheap-gap})$$

Because the endpoint gaps are $kc - o_0$ and $qc - 1 \leq 0$, the uniform condition becomes

$$o_0 \geq kc = (q - 1)\frac{\beta}{m}. \quad (\text{M-no-cheap-strong})$$

It selects exclusion for $N = 3$, and guarantees existence with exclusion for $N = 4$. For $N \geq 5$, even this stronger condition does not remove separating current passage for sufficiently high priors. Majority no-screening is therefore not an all-PBE result.

7.4 Majority boundary warning

At $\beta = 1$, or at $o_0 = 0$ with $\beta < 1$, H -yes can be optimal even when $z = q - 2$ and the current ballot still fails. Rejection classes reappear, and the regular security proof must be replaced. For $N \geq 4$, the corresponding algebraic floors are, respectively,

$$F_{M,1} = \max\{1 - kc, (1 - \mu)[1 - o_0 - (k - 1)c] + \mu c, 1 - o_1 - (k - 1)c\}, \quad (\text{M-beta-one-floor})$$

and

$$F_{M,0} = \max\{1 - kc, (1 - \mu)[1 - (k - 1)c] + \mu c, 1 - o_1 - (k - 1)c\}. \quad (\text{M-zero-low-floor})$$

These formulas are boundary bounds, not substitutes for the regular theorem. The common institutional comparison below is intentionally restricted to (regular-domain).

8 Conditional institutional comparison

8.1 Common PBE domain

Let the regular unanimity conditions in (U-existence) hold. For $N = 3$ or $N \geq 5$, this is already a common existence domain. For $N = 4$, additionally impose (M-N4-existence). These are derived existence conditions, not ad hoc primitive restrictions.

On this common domain unanimity has the single on-path payoff in (U-regular-payoffs). Majority may be set-valued, but its security value satisfies

$$F_M = \max\{E, B_M, P\} \geq P. \quad (\text{comparison-floor})$$

Combining this inequality with (M-weak-bounds) yields, in every majority PBE,

$$V_W^M \geq \frac{F_M}{m} \geq \frac{P}{m} = V_W^U. \quad (\text{entry-nesting})$$

Thus collective formation sets satisfy

$$\mathcal{F}_U \subseteq \mathcal{F}_M \quad (\text{formation-nesting})$$

selection-free on the common domain. Given a selected majority payoff $v_M = V_W^M$:

1. if $\chi \leq P/m$, both institutions form;
2. if $P/m < \chi \leq v_M$, only majority forms;
3. if $\chi > v_M$, neither forms.

There is no region in which only unanimity forms.

For H , define its expected outside payoff

$$\bar{o} = (1 - \mu)o_0 + \mu o_1.$$

Every regular majority class satisfies

$$\bar{o} \leq E[u_H^M] \leq o_1 = E[u_H^U]. \quad (\text{H-payoff-bounds})$$

Consequently, when both institutions form, unanimity weakly favors H , with equality only under majority pooling. When only majority forms, majority weakly favors H relative to nonformation under unanimity. When neither forms, H receives $\bar{o}$ under both rules. This is a conditional institutional classification over the common PBE domain, not global dominance over the primitive parameter space.

9 Reproducible verification

The analytical proofs are accompanied by separate R scripts. They check identities, cutoff order, strict and weak inequalities, named counterexamples, quota geometry, boundary formulas, and formation margins; they do not claim to prove PBE by numerical enumeration.

- `verify_clean_optout_protocol_piH0.R`: 36/36 PASS.
- `verify_clean_optout_R2_piH0.R`: 8/8 PASS.
- `verify_clean_optout_R1_piH0.R`: 11/11 PASS.
- `verify_clean_optout_majority_piH0.R`: 17/17 PASS.
- `verify_clean_optout_entry_classification_piH0.R`: 10/10 PASS.
- `verify_clean_optout_boundaries_piH0.R`: 7/7 PASS.

Machine-readable outputs are stored in `tables/`; timestamped execution logs are stored in `quality_reports/logs/`.

10 Survival matrix

Historical result	Clean immediate-opt-out status	Evidence
$\pi_H = 0$, $b_\theta = 0$, immediate o_θ Direct threshold $y_\theta^* = o_\theta$	survives as baseline primitives survives locally	Gate 0 contract and history table derived only under certain implementation and in terminal R2 $S(\nu) = \max\{P, L(\nu)\}$
Terminal unanimity P/L choice	survives	exists exactly under (U-existence)
Regular R1 pooling	survives conditionally	payment-shaving gap (U-shaving-gap)
Regular R1 low-only	disappears	survives at $\beta = 1$, not in the regular domain
R1 rejection/delay	becomes a boundary result	boundary overpay pooling and weak-PBE multiplicity add classes
Exact reduction to historical P/L/R	rejected	its hypotheses do not survive the new opt-out timing and belief discontinuity
Rejected-history reduction lemma	rejected for the clean game	unanimity and $N = 4$ majority have derived nonexistence regions
Global PBE existence over the primitive domain	rejected	large groups admit separating current passage and rents
Majority no-screening	rejected as all-PBE	uniform majority condition is (M-no-cheap-strong)
Historical No-Cheap-H $o_0 \geq \beta/m$	changes	equation (formation-nesting) on the common PBE domain
Entry nesting	survives conditionally and selection-free	equation (H-payoff-bounds) plus formation status
H institutional ranking	survives conditionally	excluded from every baseline payoff and theorem above
Delayed continuation, hybrid exit, and $t_\theta = d_\theta - b_\theta$	extension/history only	

11 Proof ledger

Object	Status	Notes
Gate -1 provenance	proved	Clean baseline commit, user-approved annotated tag, and initial v6 hash recorded.
Gate 0 extensive form	independently verified	PASS on candidate commit fff4a35 ; 36/36 protocol checks pass.
R2 unanimity	proved	Terminal low-only/pooling value and cutoff derived.
R2 majority	proved	Active and weak-only continuation values derived selection-free.
R1 unanimity, regular domain	proved; final review pending	Exact existence condition, nonexistence regions, and pooling payoff derived.
R1 unanimity, boundaries	proved or bounded as stated; final review pending	Zero outside option, no discounting, and zero terminal surplus are separated.
R1 majority, regular domain	proved; final review pending	Exact security value and $N = 3, 4, \geq 5$ characterization derived.

Object	Status	Notes
Majority boundaries	bounded; no comparison promoted	Separate floors recorded; regular theorem not extrapolated.
Entry, nesting, and classification	proved conditionally; final review pending	Selection-free on the explicitly derived common PBE domain.
Computational verification	passed; R review pending	Protocol 36/36, R2 8/8, R1 unanimity 11/11, majority 17/17, entry/classification 10/10, and boundary 7/7 checks pass.
Final formal and adversarial reviews	pending	Must review the same fixed candidate commit without editing.
Delayed continuation and other architectures	extension/history only	Not part of this Goal's clean baseline.